Student Name: _____

Student Number: _____

School: _____

Year of Entrance: _____

ShanghaiTech University Midterm Examination Cover Sheet

Academic Year: 2022 to 2023 Term: 1

Course-offering School: IMS

Instructor: *Wang, Qiang*

Course Name: **Linear Algebra I**

Course Number: MATH1112.02

Exam Instructions for Students:

1. All examination rules must be strictly observed throughout the entire test, and any form of cheating is prohibited.

2. Other than allowable materials, students taking closed-book tests must place their books, notes, tablets and any other electronic devices in places designated by the examiners.

3. Students taking open-book tests may use allowable materials authorized by the examiners. They must complete the exam independently without discussion with each other or exchange of materials.

For Marker's Use:

Section	1	2	3	4	5	6	7	8	Total
Marks									
Recheck									

Marker's Signature: Rechecker's Signature:

Date: Date:

Instructions for Examiners:

1. The format of the exam papers and answer sheets shall be determined by the school and examiners according to actual needs. All pages should be marked by the page numbers in order (except the cover page). All text should be legible, visually comfortable and easy to bind on the left side. A4 double-sided printing is recommended for the convenience of archiving (There are all-in-one printers in the university).

2. The examiners should make sure that exam questions are correct and appropriate, If errors are found in exam questions during the exam, the examiners should be responsible to respond on site, which will be taking into account in the teaching evaluation.

Specific Instructions for students:

• The time duration for this exam is 100 **minutes**.

• Computers and calculators are prohibited in the exam.

• Answers can be written in **either Chinese or English**.

★ For problems 3-7, please show details of calculations or deductions. A correct answer with no details can not earn points.

Notations and conventions:

• $\mathbb{R}$ is the set of real numbers.

• I denotes an identity matrix of suitable order.

• 0 or $\mathbf{0}$ may denote the number zero, a zero vector, or a zero matrix.

• $\mathbb{M}_{m\times n}$ is the vector space of all $m \times n$ matrices.

• For a square matrix $A = [a_{ij}]$, M_{ij} is the minor of entry a_{ij}; C_{ij} is the cofactor of entry a_{ij}.

1. Multiple choice questions.

a). (5 *points*) The scalar triple product $(\mathbf{u} \times \mathbf{v}) \bullet \mathbf{w} =$ ( )

(A) $(\mathbf{u} \times \mathbf{w}) \bullet \mathbf{v}$ (B) $(\mathbf{v} \times \mathbf{u}) \bullet \mathbf{w}$

(C) $(\mathbf{w} \times \mathbf{v}) \bullet \mathbf{u}$ (D) $(\mathbf{v} \times \mathbf{w}) \bullet \mathbf{u}$

b). (5 *points*) Which of the following is linearly independent? ( )

(A) $\left\{[1, 2, 3, 4]^T, [2, 3, 4, 5]^T, [3, 4, 5, 6]^T, [0, 0, 0, 0]^T\right\}$

(B) $\left\{[1, 2, -1]^T, [3, 5, 6]^T, [0, 7, 9]^T, [1, 0, 2]^T\right\}$

(C) $\left\{[a, 1, 2, 3]^T, [b, 1, 2, 3]^T, [c, 3, 4, 5]^T, [d, 0, 0, 0]^T\right\}$

(D) $\left\{[a, 1, b, 0, 0]^T, [c, 0, d, 0, 0]^T, [a, 0, c, 5, 6]^T\right\}$

c). (5 *points*) The determinant $\begin{vmatrix} 0 & a & b & 0 \\ a & 0 & 0 & b \\ 0 & c & d & 0 \\ c & 0 & 0 & d \end{vmatrix} =$ ( )

(A) $(ad - bc)^2$ (B) $-(ad - bc)^2$

(C) $a^2d^2 - b^2c^2$ (D) $b^2c^2 - a^2d^2$

2. Fill in the blanks.

a.) (5 *points*) Let $\mathbf{u} = [3, -2, 1]^T$, $\mathbf{v} = [-1, k, -5]^T$. If $||proj_{\mathbf{u}}\mathbf{v}|| = \sqrt{14}$, then k=_____.

b). (5 *points*) If every vectors in $\mathbb{R}^3$ can be expressed as a linear combination of the following three vectors:

$$\mathbf{v}_1 = [1, 4, 2]^T, \mathbf{v}_2 = [2, 7, 3]^T, \mathbf{v}_3 = [0, 1, a]^T$$

then $a =$ _____.

3. (10 *points*) Let $\mathbf{u} = [1,\ 1,\ 2]^T$, $\mathbf{v} = [0,\ 2,\ 3]^T$ in $\mathbb{R}^3$. Suppose that H is a plane in $\mathbb{R}^3$ that passes through the point $P(1, 0, 1)$, and is parallel to both $\mathbf{u}$ and $\mathbf{v}$. Find the equation of H in the *point-norm form.*

4. (10 *points*) Let $A = \begin{bmatrix} 1 & 1 \\ -1 & 0 \\ 0 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 2 & -1 \\ 2 & 2 & -1 \end{bmatrix}$. Evaluate $(AB)^{2022}$.

5. (10 *points*) Find the inverse of the matrix: $A = \begin{bmatrix} 1 & 3 & -5 & 7 \\ 0 & 1 & 2 & -3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

6. (10 *points*) Suppose A is a 4×4 matrix, the first row of A is $[2, 4, 1, -x]$, and the cofactors of the third row of A are $C_{31} = 1$, $C_{32} = x^2$, $C_{33} = -4x$, $C_{34} = 2x$. Compute x.

7. (15 *points*) Given the linear system

$$\begin{cases} (2-\lambda)\,x_1 + 2x_2 - 2x_3 = 1 \\ 2x_1 + (5-\lambda)x_2 - 4x_3 = 2 \\ -2x_1 - 4x_2 + (5-\lambda)x_3 = -\lambda - 1 \end{cases}$$

solve the following problems.

a). Determine the value of λ so that this system is inconsistent.

b). Find the value of λ so that the system has a unique solution.

c). Determine λ so that the system has infinitely many solutions and find the general solution.

8. (20 *points*) Suppose $\{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n\}$ is a basis for V.

a). Can $\{\mathbf{v}_1,\ \mathbf{v}_1 + \mathbf{v}_2,\ \cdots,\ \mathbf{v}_1 + \cdots + \mathbf{v}_n\}$ be a basis for V? If not, please explain; if yes, please give proof.

b). Can $\{\mathbf{v}_1 + \mathbf{v}_2,\ \mathbf{v}_2 + \mathbf{v}_3,\ \cdots,\ \mathbf{v}_{n-1} + \mathbf{v}_n, \mathbf{v}_n + \mathbf{v}_1\}$ be a basis for V? If not, please explain; if yes, please give proof.